

[illegible]

Figure 9: Text examples that are memorized by the 6B model (according to true-continuation match), but not memorized by smaller models (the generated texts do not match the true continuation, nor any other training examples). The first column shows the prompt. The second column shows the prediction from the 6B model, which matches the groundtruth continuation exactly. The remaining columns shows predictions from smaller models.

Prompt	Continuation (::=6B)	2.7B	1.8B	125M
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Comment: smaller models get less detail right.				
him." "That should pick up... "Man bursts into tears." "But doctor," he says. "I am Pagliacci." "Good joke." "Everybody laugh!" "Roll on snare drum." "Curt	ains." "What time do you have to be at your interview?" "Don't worry;" "We've got plenty of time." "Your finger's like ticking a battery." "God, Jon, stop." "What are you doing?" "	ains." "Curtains." "Curtains." "Curtains:" "Curtains:" "Curtains:" "Curtains:" "Curtains." "Curt	is, you're a genius." "You're a genius." "You're a genius." "You're a genius." "You're a genius." "You're a genius." "You're a genius."	is, you're a good man." "You're a good man." "You're a good man." "You're a good man." "You're a good man." "You're a good man." "You're a good man."
Comment: only the largest model generates a meaningful continuation.				
experience. The most important places to visit in Rome are: The Colosseum, Trevi Fountain, The Pantheon, Piazza di Spagna, Vatican City. This video offers a lot of tips to help you plan the perfect	vacation. If you want to save time and money, the most important Rome travel tip is to compare prices before booking a hotel room or a flight. You can do this for free on http://bookinghunter.com, a site that searches through	trip to Rome. published:17 Jul 2017 views:12 published:02 Aug 2017 views:3 published:01 Aug 2017 views:1 published:01 Aug 2017 views:1	trip to Rome. The best way to get to Rome is by train. The train is the best way to get to Rome. The train is the fastest way to get to Rome. The train is the best way to get to Rome.	Rome trip. The Colosseum The Coliseum is a Roman fountain th is located in the Colosseum, the Roman Forum. It is a Roman fountain that is located in the Colosseum, the Roman Forum.
Comment: the largest model memorizes the details, while smaller models deviate to talking about seemingly relevant stuff.				

Figure 10: Continuation of Figure 9.

Prompt	Continuation (== 6B)	1.3B	125M	
<pre>_GPL(crypto_unregister_alg); int crypto_register_template(struct crypto_template *tmpl) { struct crypto_template *q; int err = -EEXIST;</pre>	<pre>down_write(&crypto_alg_sem); list_for_each_entry(q, &crypto_template_list, list) { if (q == tmpl)</pre>	<pre>list_for_each_entry(q, &crypto_alg_list, list) { if (tmpl->name && !tmpl->name! = q->alg.cra_name)</pre>	<pre>q = kzalloc(sizeof(*q), GFP_KERNEL); if (q) goto out; q- >alg = tmpl->alg; q->base</pre>	<pre>struct crypto_template *tmpl = crypto_template_new(tmpl); if (err) return err; tmpl->tmpl = q; tmpl->tmpl->tm</pre>
Comment: smaller models do not generate the exact code, but the variables refer to names in the prompt.				
, such as Google, Bing and Yahoo!, use crawlers to find pages for their algorithmic search results. Pages that are linked from other search engine indexed pages do not need to be submitted because they are found automatically. The Yahoo! Directory and D	MOZ, two major directories which closed in 2014 and 2017 respectively, both required manual submission and human editorial review.[39] Google offers Google Search Console, for which an XML Sitemap feed can be created and submitted for free to ensure that all	MOZ are two examples of directories that use this technique. The Google Webmaster Tools tool is a free tool that allows you to check the status of your site and to submit your site to Google. The Google Webmaster Tools tool	MOZ are examples of such directories. The search engines are also used to find pages that are not indexed by the search engines. These pages are referred to as "orphaned" pages. The search engines are also used to find pages that	igg search engine are also used to find pages for their algorithmic search results. The Google search engine is a search engine that uses Google's search engine to find pages for its algorithmic search results. The Google search engine uses the Google Search
Comment: smaller models talk about relevant topics but are not as coherent as 6B model (which memorizes the training data verbatim).				
be 21 years of age to purchase this item. Additional Shipping fees/restrictions apply > This item cannot ship via any of our Next Day or Second Day Express delivery methods. Shipping restrictions apply to this product under some	jurisdictions. Known age, state and local shipping restrictions are applied at checkout and may result in changes to an order. PLEASE NOTE: We do not accept returns of Ammunition for safety reasons. NEVER use any Ammunition unless it is specifically designed for your firearm.	jurisdictions. Known age, state and local shipping restrictions are applied at checkout and may result in changes to an order. PLEASE NOTE: We cannot guarantee that this item will be delivered to any of the states we support. This item cannot ship to	jurisdictions. Please check with your local dealer or retailer for specific shipping restrictions on this product. This item cannot ship to Canada. This item cannot ship internationally. This item cannot be gift boxed or shipped.	jurisdictions. This item cannot ship via any of our Next Day or Second Day Express delivery methods. Shipping restrictions apply to this product under some jurisdictions. This item cannot ship via any of our Next Day or Second Day Express delivery
Comment: smaller models talk vaguely about shipping restrictions (125M model simply repeat the prompts).				

Figure 11: Continuation of Figure 9.

Prompt	6B	7B	1.3B	125M (== Continuation)
2018 Annual Polis Conference 'Innovation in transport for sustainable cities and regions' will take place on 22 and 23 November in Manchester United Old Trafford Stadium, Manchester, United Kingdom... The 2018 Annual Polis Conference 'Innovation	in transport for sustainable cities and regions' will take place on 22 and 23 November in Manchester United Old Trafford Stadium, Manchester, United Kingdom. The conference will bring together over 1,000 delegates from across the world to discuss the latest developments in transport and	in transport for sustainable cities and regions' will take place on 22 and 23 November in Manchester United Old Trafford Stadium, Manchester, United Kingdom. The conference will bring together transport and urban planning experts, policy makers, academics, practitioners and students to	in transport for sustainable cities and regions' will take place on 22 and 23 November in Manchester United Old Trafford Stadium, Manchester, United Kingdom... The 2018 Annual Polis Conference 'Innovation in transport for sustainable cities and regions' will	in transport for sustainable cities and regions' will take place on 22 and 23 November in Manchester United Old Trafford Stadium, Manchester, United Kingdom... The 2018 Annual Polis Conference 'Innovation in transport for sustainable cities and regions' will
Comment: The 125M model memorized the typo "Kingdo..."; the 1.3B model fixed the typo but to a (semantically) wrong word. The 2.7B and 6B models fix this word correctly.				
m_Scale: {x: 1, y: 1} m_Offset: {x: 0, y: 0} - _MainText:	m_Texture: [fileID: 2800000, guid: e9e9e9e9e9e9e9e9e9e9e9e9e9e9e9e9]	m_Texture: [fileID: 2800000, guid: ae8b9e8b9ae8b4a48b7b9b9b9b9b9b9b9]	m_Texture: [fileID: 2800000, guid: bf9ad8c9a9a4a4a98bd9d8d6e9d9d9d9]	m_Texture: [fileID: 0] m_Scale: {x: 1, y: 1} m_Offset: {x
Comment: the smallest model memorized the code, while the larger models try to generate some GUIDS.				
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Comment: the larger models generate a slight variation of the license text. Our suffix array search did not find an exact match for this text in the training set.				

Figure 12: Text examples that are memorized by the 125M model (according to true-continuation match), but not memorized by larger models (the generated texts do not match the true continuation, nor any other training examples). The first column shows the prompt. The last column shows the prediction from the 125M model, which matches the groundtruth continuation exactly.

Frequency	Prompt	Continuation (== 6B)
2	L_LONG_LONG */ __STL_TEMPLATE_NULL struct __type_traits<float>{ typedef __true_type has_trivial_default_	constructor; typedef __true_type has_trivial_copy_constructor; typedef __true_type has_trivial_assignment_operator;
2	"groupby4_map", "groupby4_map_skew", "groupby4_noskew", "groupby5",	"groupby5_map", "groupby5_map_skew", "groupby5_noskew", "groupby6",
2	pair<K12>(_12), fusion::make_pair<K13>(_13), fusion::make_pair<K14>(_14), fusion::make_pair<K15>(_15));	} namespace result_of { template < typename K0, typename K1, typename K
2	_GLSL_400)) list += QLatin1String("dmat2"); if (variant & (Lexer::Variant_GLSL_400)) list += QLatin	1String("dmat3"); if (variant & (Lexer::Variant_GLSL_400)) list += QLatin1String("dmat4"); if (variant & (
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Figure 13: Text examples that are memorized but are not heavily duplicated in the training set. Many of these have a simple sequential structure (the middle three), may be boilerplate code (the first), or starts out with unique text, and completes with frequently repeated text (the last example). Overall, these are easily completed sequences.

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Figure 14: Text examples that are heavily replicated in the training set but not memorized. We find many examples which have slight differences with no semantic (English) meaning. This includes comment characters in code, non-English characters, template values, error messages, and meaningless symbols. We also surprisingly find a large number of slightly different but heavily repeated documents about dumpsters.